FORMULAS AND IDENTITIES

Algebraic Square Formulae

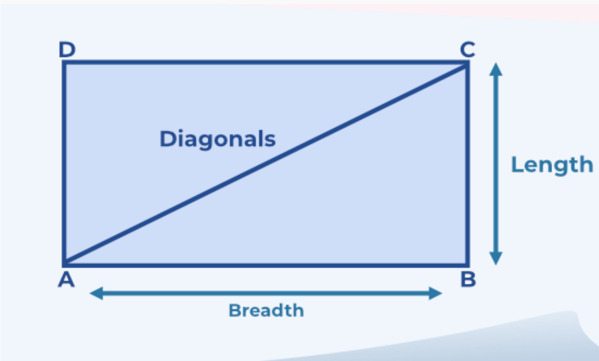
1. $a^2 - b^2 = (a+b)(a-b)$
2. $(a+b)^2 = a^2 + b^2 + 2ab$
3. $(a-b)^2 = a^2 + b^2 - 2ab$
4. $ab = \frac{(a+b)^2 - (a^2+b^2)}{2}$
5. $2(a^2 + b^2) = (a+b)^2 + (a-b)^2$
6. $4ab = (a+b)^2 - (a-b)^2$
7. $(a+b)^2 = (a-b)^2 + 4ab$
8. $(a-b)^2 = (a+b)^2 - 4ab$
9. $a^2 + b^2 = (a+b)^2 - 2ab = (a-b)^2 + 2ab$

Algebraic Cube Formulae

1. $(a+b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$
2. $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$
3. $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$
4. $a^3 + b^3 = (a+b)[(a+b)^2 - 3ab]$
5. $a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$
8. $(a-b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$
9. $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
10. $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$
11. $a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$
12. $a^3 - b^3 = (a-b)[(a-b)^2 + 3ab]$

Rectangle Formulae

1. Area: $A = l \times w$
2. Perimeter: $P = 2(l + w)$
3. Diagonal: $d = \sqrt{l^2 + w^2}$



Square Formulae

1. Area: $A = a^2$
2. Perimeter: $P = 4a$
3. Diagonal: $d = a\sqrt{2}$